

FedMAQ: Multi-Adaptive Quantization & Knowledge Distillation

Communication-Efficient Federated Learning

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- We shifted focus from physical IoT/Smart Campus energy forecasting to classical Federated Learning.
- We chose image classification (e.g., MNIST, CIFAR) to make experiments more manageable and reproducibility easier.
- We want to explore multi-adaptive quantization with server-side knowledge distillation with the hopes of further reducing communication while maintaining comparable accuracy.

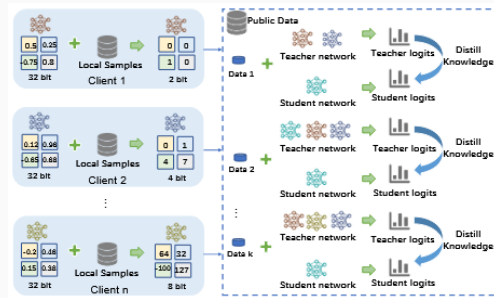
(Some) Related Works on Communication-Efficient FL

Algorithm	Time-Aware	Data-Aware	Resource-Aware	Uses KD	Key Components
DynFed	Yes	Yes	No	Yes	Quantization-Aware KD
AdaDQ-KD	No	No	Yes	Yes	Dithering Quantization & Diff. Privacy
DAdaQuant	Yes	Yes	No	No	Doubly-Adaptive Fixed-Point
FedDT	No	No	No	Yes	Ternary Compression
LAQ-HC	No	Yes	Yes	No	Reinforcement Learning
FedPAQ	No	No	No	No	Periodic Averaging
FedMAQ (Ours)	Yes	Yes	Yes	Yes	3D-Awareness + Logit KD

We explore considering client resource constraints, data skew, and temporal training dynamics.

FedMAQ Architecture

- The server broadcasts the global model parameters to participating client nodes.
- Clients perform local optimization steps on their non-IID datasets.
- Uplinks are compressed using multi-adaptive quantization based on live bandwidth and stability.
- The server aggregates client logits via knowledge distillation to neutralize quantization noise.

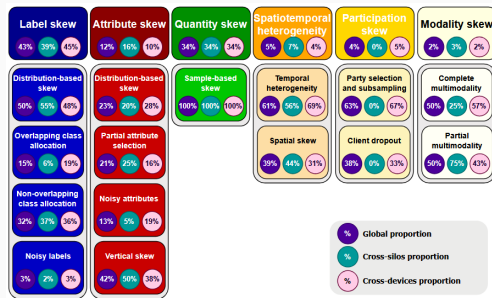


He et al. (2025 - DynFed), Figure 2: Overview of the training procedure using logits.

Our work is a logical extension of DynFed by He et al (2025).

Experimental Setup

- Five standard vision benchmarks are selected: MNIST, FMNIST, CIFAR-10, CIFAR-100, and FEMNIST.
- Data distributions across clients are skewed using a Dirichlet concentration parameter $\alpha \in \{0.1, 1, 10\}$.
- Bandwidth bottlenecks are simulated using a uniform distribution $U(5, 20)$ Mbps.



Jimenez et al. (2024), Figure 7: Taxonomy of different types of skew and heterogeneity.

Statistical and system heterogeneity are modeled simultaneously to reflect realistic edge environments.

Classification of Communication-Efficient FL Related Works

Foundational Baselines

FedAvg, FedProx

Static Compression

DAdaQuant, FedPAQ

Pure Distillation

FedKD, FedDistill

Adaptive Compression

AdaGQ, LAQ-HC

Direct SOTA Competitors

DynFed, AdaDQ-KD, FedDT

An 11-algorithm evaluation stack benchmarks FedMAQ against compression, distillation, and adaptive SOTA.

Manuscript Progress

Chapter	Status & Accomplishments
Chapter 1: Introduction	Outlined
Chapter 2: Related Works	Outlined
Chapter 3: Theoretical Framework	Outlined
Chapter 4: Methodology	Outlined

Next Meeting Commitments

- Configure the Flower-based Federated Learning simulation environment for Phase 1 baselines.
- Implement Dirichlet non-IID partitioning pipelines across the selected image classification datasets.

Thank You!

Questions & Answers